

Comparison of Erasure-Aware Error Correction Decoding for Fault Tolerant Quantum Computing

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C191A Introduction to Quantum Computing I
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Abstract

TODO

The abstract should summarize our main points: purpose, methods, results, and conclusions.
Keep it short.

Contents

1 Introduction

Motivation & research question

Motivation: Fault-tolerate quantum computation at scale remains a formidable challenge, but we’re excited about these recent major advancements:

Decoding with erasure information dramatically increases the surface code error threshold from 0.937% to 4.15%, with an estimated 98% of errors convertible into erasures. [wu_erasure_2022-1](#)

Machine-learned error decoding for quantum processors (surface code) by Bausch et al uses a transformer-based neural network to decode error data with realistic noise models, including qubit cross-talk and state leakage.

"Requires a system capable of dissipatively removing physical errors while simultaneously performing arbitrary coherent manipulation of the encoded logical qubits." [bluvstein_architectural_2025](#)

Sub-questions & hypotheses

2 Background & Related Work

2.1 Noise Models

Pauli Noise. The simplest model used to evaluate Quantum Error Correction (QEC) assumes all physical qubit errors can be represented by uniformly random Pauli operators applied during circuit simulation. In our simulations we implement a [depolarizing error channel](#) to represent [Pauli noise](#), which has the following effect on some quantum state ρ with p probability of becoming a maximally-mixed state and $(1 - p)$ probability of remaining unchanged:

$$E_{\text{pauli}}(\rho) = (1 - p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z) \quad [\text{Depolarizing error channel}] \quad (1)$$

Unfortunately, error models based only on [Pauli noise](#) are not robust enough to describe the real-world noise that must be corrected to realize fault-tolerant quantum computation. [CITATION NEEDED]

Leakage. A [leakage error](#) occurs when a qubit “leaks” out of the computational subspace (described by a two-dimensional Hilbert space) into a state that can only be described with a higher-order Hilbert space. If leakage goes undetected, the error compounds over time as it spreads through a multi-qubit interactions. Leakage can be stochastically modeled with a 3-level system (qutrit) with orthonormal basis: $\{|0\rangle, |1\rangle, |2\rangle\}$. Leakage events are assumed to be discrete and occur independently with probability p_{leak} at each gate operation acting on the qutrit [suchara_leakage_2014](#). The leakage error $E_{E_{\uparrow}}$ on the j th qutrit in the system can then be understood as:

$$E_{\text{leakage}}(\rho_j) = (1 - p_{\text{leak}})\rho_j + p_{\text{leak}}|2\rangle\langle 2| \quad [\text{Leakage error channel } \text{suchara_leakage_2014}] \quad (2)$$

Leakage-reduction units (LRUs) were proposed as a way to convert leakage errors into Pauli-type errors, based on a one-bit teleportation protocol that effectively moves the logical state from the physical qubit in the leaked state to a freshly-prepared physical qubit [suchara_leakage_2014](#).

All species of qubit experience leakage errors, but it’s important to note the underlying *physical* processes look and behave very differently despite sharing a common mathematical description.

Transmon leakage. Transmon single and two qubit gates rely on coherent microwave driving to transition between computational basis states. Two qubit gates such as the controll Z and controll

X gates are the primary source of leakage in transmon qubits because of their temporary usage of states outside of the computational basis to induce phase and bit flips in the target qubit. For example, when applying a control Z gate on transmon hardware to the state $|11\rangle$, we temporarily force both excitons onto one qubit, forcing it into the second excited state, $|2\rangle$, and letting the other qubit fall into the ground state, $|0\rangle$, before allowing the system to relax back into $|11\rangle$ with the intended phase shift applied. However, there is a nonzero chance that the system does not relax properly, and leaves one of our qubits in the $|2\rangle$ state, causing a leakage error in our system.

Yang_2024, Miao_2023

Neutral atom / Rydberg leakage. Neutral atom entangling gates exploit Rydberg blockade, which is a quantum phenomenon that occurs when an atom is excited to a particular Rydberg energy level, which prevents nearby atoms from being excited to the same state. The excitation of the single valence electron in alkali metals like Cs, Rb results in a large effective atomic radius [CITATION NEEDED, but I think it's on the order of $25E5$], creating a large electric dipole moment.

(one or more electrons in the case of alkaline earth metals or lanthanides like Yb) **TODO morgado_quantum_2021**

Erasure. An [erasure error](#) occurs when a qubit is longer detectable by the measurement device, so the quantum state information encoded by the qubit is effectively “lost” or “erased”. Mathematically, we extend the qubit Hilbert space with some $|e\rangle$ that is orthogonal to the qubit state $\{|0\rangle, |1\rangle\}$. The erasure channel can be defined as:

$$E_{\text{erase}}(\rho_j) = (1 - p_{\text{erase}})\rho_j + p_{\text{erase}} |e\rangle \langle e| \quad [\text{Erasure error channel } \textbf{suchara_leakage_2014}] \quad (3)$$

Photon loss is the dominant form of error in photonic qubits [CITATION NEEDED].

Leakage-detection units (LDUs) were developed to address leakage errors in superconducting qubits, by periodically coupling a data qubit to an ancilla that flags when the data qubit leaves the computational subspace. The flagged data qubit can be reset/replaced during subsequent error-correction cycles, effectively treating the flagged qubit as an erasure error.

Neutral atoms / Rydberg arrays, each atom is arranged into a known position using optical tweezers. Although the lattice/trap is situated inside a vacuum chamber, some nonzero amount of background gas collisions or thermal energy transfer from the environment can eject an atom from its lattice site.

Correlated Errors. **TODO** The most important thing to mention is that real-world errors are often correlated, so fault-tolerant QC requires correction of correlated noise. **wilen_correlated_2021**

Cross-talk might also be worth mentioning for completeness.

It is common and convenient to assume for general quantum error correcting codes that errors occur independently from each other. This assumption simplifies the decoding process but does not accurately reflect the behavior of the real world. In practice, quantum noise is often correlated. These correlated errors can cause effects not handled by independent-noise models and complicate the decoding process.

Particularly, in neutral atoms, there exists the chance of cross-talk between atoms when performing two qubit gates. Rydberg blockade is a necessary step to entangle two atoms, but applying the Rydberg excitation laser can unintentionally excite neighboring atoms

Minimum Weight Perfect Matching (MWPM), which is traditionally used for surface codes, can only handle pairwise correlations. This means that the decoder fails when a single error involves more than 2 qubits. However, for MWPF, since it changes the MLE problem into an equivalent hypergraph problem, the hypergraph can capture multi-qubit errors through its hyperedges, and thus can effectively represent correlated errors on more than 2 qubits.

The existence of correlated errors further motivates the use of MWPF over MWPM, as its hypergraph representation effectively captures correlated errors into its hyperedges, something which MWPM cannot do.

2.2 Erasure Qubits

A qubit engineered so the dominant source of errors is from erasures.

A bias towards erasure-type errors occurs natively in some types of qubits (e.g. photons) or can be engineered by converting leakage errors to erasures (e.g. with LDUs in superconducting qubits).

2.2.1 Advantages, Challenges of Erasure Errors

Advantages. From: "The benefits and costs of quantum error correction with erasure qubits" (QIP 2025)

1. Can theoretically correct more erasure errors than unknown Pauli errors. We can correct $d - 1$ erasure errors compared to $\frac{d-1}{2}$ Pauli errors. As a result, we expect logical error rate P_L on the order of p^d where p is the physical qubit error rate, compared to $p^{d/2}$ for unknown Pauli errors. **gu_2023**
2. Higher thresholds: surface code 50% erasure noise vs. 19% depolarizing noise.
3. Dominant noise source can be engineered

TODO Figure: $d=5$ surface code with some unknown Pauli error spanning 3 qubits, with syndrome squares highlighted. If we apply the minimum-weight correction, this causes a logical error.

Figure: $d=5$ surface code with erasure errors, exact locations known.

Challenges. (TODO: Add sources, possibly point 4?) Challenges from "The benefits and costs of quantum error correction with erasure qubits" (QIP 2025):

1. More physical operations.
2. Erasure detection imperfect, introducing another dimension to error models.
3. Midcircuit erasure checks can create correlated noise for neighboring qubits
4. Tools to simulate and decode erasure circuits are active area of research.

In a neutral atom, erasure is a phenomenon that can occur due to atoms leaving their optical traps or when atoms enter an energy level that is known but outside of the computational subspace. Unlike Pauli errors, the locations of the erasure errors can be known, but cannot be solved by simply applying a few logic gates. Especially when the atom is lost.

While erasure errors have a myriad of benefits, they also prove to bring some challenges:

Erasure errors often require extra physical operations. Erasure checks consist of checking if a qubit is in a known, non-computational subspace, and if it is determined that the atom has experienced an erasure error, then erasure resets will bring qubits who are in a known, non-computational subspace back into a computational space. Whether the atom has been lost, or whether it has entered an invalid energy level will determine how the erasure reset mechanism functions.

Another difficulty that presents itself with erasure errors is that they can spread through entangling operations as Pauli errors. If a qubit has experienced an erasure error but it has not yet been detected by erasure checks, the error will propagate onto neighboring qubits. What makes this particularly difficult to deal with is this turns the errors from being a known erasure error into correlated unknown errors.

In principle, one could perform erasure checks before every single gate operation to prevent the error spreading, but this method is far more expensive. Erasure checks are also prone to adding noise. It is thus necessary to achieve a balance between checking frequently enough to prevent the spread of errors, while also taking into account the cost of checking frequently and the risks it carries.

Erasure Conversion in Superconducting Qubits

1. Shouzhen Gu. "Fault-tolerant quantum architectures based on erasure qubits" and "Optimizing quantum error correction protocols with erasure qubits" **gu_optimizing_2024**
gu_fault-tolerant_2025

Erasure Conversion in Neutral / Rydberg Atoms

1. Yue Wu, Shimon Kolkowitz, Shruti Puri, Jeff D Thompson. Erasure conversion for fault-tolerant quantum computing in alkaline earth Rydberg atom arrays **thompson_erasure_2022**
2. Gefen Baranes, Madelyn Cain, J. Pablo Bonilla Ataides, Dolev Bluvstein, Josiah Sinclair, Vladan Vuletic, Hengyun Zhou, Mikhail D. Lukin . Leveraging Atom Loss Errors in Fault Tolerant Quantum Algorithms **baranes_leveraging_2025**

2.2.2 Formalism

define stabilizer circuits with erasure channel: **gu_fault-tolerant_2025**
compare w/ surface code having only pauli error channel

3 Methods

Simulating Erasure Circuits Describe Stim circuits. Link to Stim circuits / appendix figures of selected timeline-svg operations. **gidney_stim_2021**

Decoding TODO

Describe the decoding problem for Pauli error stabilizer circuits in the surface code. Explain how surface code error syndromes can be framed as a graph traversal problem with detection events as nodes and errors as the weighted edges. The minimum weight perfect matching (MWPM) algorithm can find the most probable set of errors that occurred. Explain why MWPM works for uniformly random Pauli errors, but why it experiences a sharp drop-off in performance for biased (non-identical) noise models.

Describe the decoding problem when 100 of noise channels can be converted to erasure errors (conversion_ratio=1 in our codebase). Describe how the Hypergraph Minimum-Weight Parity Factor (MWPF). **wu_minimum-weight_2025**

Minimum Weight Parity Factor (MWPF) provides a hypergraph-based reformulation of the Most Likely Error (MLE) decoding problem. The purpose of MLE is to find the lowest-weight error configuration consistent with a given syndrome. However, the number of error configurations grows exponentially with the number of qubits, making MLE intractable; in fact, the problem is known to be NP-hard.

MWPF maps the MLE problem into a general hypergraph problem, retaining its intractability and NP-hardness. Although the conversion does not solve the NP-hardness of MLE, the hypergraph conversion allows us to employ methods to approximate a near optimal solution in approximately polynomial time. To employ this conversion efficiently, MWPF uses Hyperblossom, a generalization of the blossom algorithm for Minimum Weight Perfect Matching (MWPM) to a hypergraph equivalent. Hyperblossom is constructed with a primal-dual linear programming (PDLP) framework, which introduces relaxation and transformations to the hypergraph.

PDLP offers two major advantages. It allows for a certifiable proximity bound, allowing us to quantify how close the approximate solution is to the true minimum-weight parity factor. Second, given certain simple hypergraphs, it can find the exact minimum-weight parity factor solution in polynomial time.

Clustering and Relaxing are the methods used to approximate the NP-hardness of the MWPF into something that can be solved in approximately polynomial time. Clustering exploits the locality of the errors by breaking the hypergraph into subgraphs. Some subgraphs, called relaxers, can be solved exactly in polynomial time. The other subgraphs which are not exactly solvable are handled by Singlehair, which assigns edges/variables in a way that's consistent with the syndrome. As such, combining all the local solutions into one global error estimate. This scales linearly with the number of qubits, so the entire process can be done in polynomial time in the average case.

To address the NP-hardness of the hypergraph, MWPF incorporates two techniques: clustering, and relaxation. Clustering exploits the locality of the errors by breaking the hypergraph into smaller subgraphs. Certain subgraphs, called relaxers, can be solved in polynomial time through Hyperblossom due to their simplicity. Subgraphs which cannot be solved exactly are handled using SingleHair, a process which assigns variables and hyperedges in a syndrome consistent manner. Local solutions from each cluster are combined into one global error estimate. Because the process breaks the decoding task into subproblems that scale with the locality of the errors rather than the number of qubits, the overall algorithm scales linearly with the number of qubits and therefore achieves polynomial-time in the average case.

MWPF stands out from the rest of the decoders. First, many of these decoders are limited to surface codes. Since we introduce erasure errors, our problem turns into a hypergraph problem which is perfect for MWPF but cannot be handled by other decoders. Second, thanks to PDLP, MWPF offers certifiable proximity bound. Its use of clusters and relaxers achieve an approximate polynomial-time solution to what would otherwise be a NP-hard problem.

4 Metrics

5 Results

6 Discussion

7 Conclusion

Acknowledgements Shimon Kolkowitz, Harry Levine, Thiago Bergamaschi

A Appendix: Syndrome of Loss Errors

Describe how a loss is heralded during state readout. Each potential loss location is described by a distinct error probability, described by a unique configuration of the error syndroms. [EXAMPLE OF ERROR SYNDROME]

B Appendix: Additional Figures and Tables

